SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Deep Learning COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

Duration: 1 Hour

QUESTIONS: 5 Total Marks:50

Annexure A

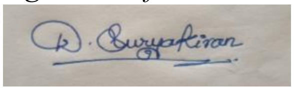
Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:
   H H T H H H T H H T
   Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).
2. Out of 50 students, 42 passed an exam.
   Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.
3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective. Find the MLE for the probability that a bulb is defective.
4. A biased die is rolled 60 times. The number 6 appears 18 times.
   Estimate the probability of rolling a 6 using MLE.
5. Given: Input (x) = 4
   Weight (w) = 2
   Actual Output =
   10 Prediction: y =
   w × x
   1. Find the predicted output.
   2. Calculate the error.
   3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Code of Conduct:

I D.Suryakiran(192325057) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Answers

1. A coin is tossed 10 times and the outcomes are:

H H T H H H T H H T?

Given the outcomes:

H H T H H H T H H T

Step 1: Count the outcomes

- Total tosses n=10n = 10n=10
- Number of Heads H=7H = 7H=7
- Number of Tails T=3T = 3T=3

Step 2: MLE formula

For a coin, the Maximum Likelihood Estimate of the probability of Heads is:

p^=Number of Head\Total Tosses p^=7\10=0.7

2. Out of 50 students, 42 passed an exam. Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

Given:

- Total students = 50
- Students who passed = 42
- Students who failed = 50−42=850-42=850−42=8

MLE Formula

For a Bernoulli distribution, the MLE of probability of success is:

P^=Number of successes\Total Observations

Here, **passing = success**.

P^=42/50

P^=0.84

3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective. Find the MLE for the probability that a bulb is defective.

Given:

- Total bulbs = 100
- Defective bulbs = 8
- Non-defective bulbs = 92

MLE Formula

For a Bernoulli distribution:

p^=Number of defective bulbs\Total bulbs

Substitute the values:

p^=8/100

p^=0.08

4. A biased die is rolled 60 times. The number 6 appears 18 times.

Estimate the probability of rolling a 6 using MLE.

Given:

- Total die rolls = 60
- Number of times 6 appears = 18 MLE Formula p^=Number of successes\Total trials Here, rolling a 6 is considered a success.

p^=18\60 p^=0.3

5. Given:

Input (x) = 4

Weight (w) = 2 Actual

Output = 10

Prediction: y

= w × x

1. **Find the predicted output.**
2. **Calculate the error.**
3. **If the gradient is 4 and learning rate is 0.01, find the updated weight.**

Given:

- Input x=4x = 4x=4
- Weight w=2w = 2w=2
- Actual output =10=10=10
- Gradient =4=4=4
- Learning rate η=0.01\eta =0.01η=0.01 1. Predicted Output

Formula: $y=w\times x$

$y=2\times 4=8$

So, Predicted Output = 8.

2. Calculate the Error Using:

Error=Actual−Predicted

Error=10−8=2 So,

Error = 2.

3. Updated Weight

Gradient descent formula:

$Wnew = wold - (\text{Learning Rate} \times \text{Gradient})$wRate\times Gradient)
$Wnew = 2 - (0.01 \times 4)$

$= 2 - 0.04$

$Wnew = 1.96$